

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Architected and shipped end-to-end **ReactJS** data applications with **rich user interfaces using HTML5 and CSS3**, integrating **RESTful service layer components in Java and Python** to deliver business-critical metrics dashboards with sub-second query latency across **relational databases (PostgreSQL, MySQL, Oracle)**, reducing insight delivery time by **60%**.
- Designed and implemented **NodeJS** backend services powering interactive data visualizations, leveraging **test-driven development** practices and **Software Engineering processes** to ensure **99.5% application uptime** while supporting concurrent access by 500+ internal stakeholders across distributed teams.
- Developed production-grade data applications combining **open source technologies** with internal infrastructure, demonstrating advanced **problem-solving and critical thinking** to evaluate and apply emerging technologies in short timeframes, achieving **40% reduction in deployment cycle time** through optimized CI/CD pipelines.
- Owned complete application development lifecycle from concept through production deployment, partnering with internal stakeholders to understand business workflows and champion platform enhancements that transformed user experience; delivered features on consistent release schedule with **100% on-time delivery** across three consecutive quarters.
- Applied **data structures and algorithms** expertise to optimize application performance, implementing caching strategies and query optimization across relational databases that improved dashboard load times by **45%** and reduced database CPU utilization by **35%**.
- Integrated **Generative AI** and **Large Language Model** capabilities into data applications, implementing **prompt engineering** techniques to deliver AI-powered insights that enhanced analyst productivity by **50%**, while maintaining rigorous code quality through peer reviews and automated testing frameworks.

SKILLS

Frontend & UI Development – ReactJS, HTML5, CSS3, AngularJS, Vue.js, interactive data visualization, responsive design.

Backend & API Development – Java Spring Boot, Python, NodeJS, RESTful service layer components, FastAPI, Flask.

Databases & Data Structures – PostgreSQL, MySQL, Oracle, relational data modeling, SQL query optimization, data structures, algorithms.

Software Engineering & Methodology – test-driven development, CI/CD (Jenkins), Git, code review, Agile/Scrum, Software Engineering processes, change management.

AI & Data Technologies – Generative AI, Large Language Models, prompt engineering, data analysis, machine learning (scikit-learn, pandas, NumPy).

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, application monitoring, production support.

Analytics & Business Intelligence – Power BI, Tableau, DAX, Power Query, data warehousing, ETL, business intelligence dashboards.

EDUCATION AND TRAINING

PROJECTS

AI-Powered Analytics Data Application Platform (HCL Technologies)

- Architected a comprehensive **ReactJS** and **NodeJS**-based data application platform delivering real-time business metrics with **Large Language Model** integration, enabling business partners to query complex datasets using natural language; reduced query formulation time from 15 minutes to 30 seconds through **prompt engineering** techniques.
- Designed **RESTful service layer components in Java Spring Boot** and **Python** backends interfacing with **PostgreSQL and Oracle** relational databases, implementing advanced **data structures and algorithms** for query optimization; achieved **sub-300ms response times** on analytics queries serving 1,000+ concurrent users.
- Developed rich, interactive user interfaces with **HTML5 and CSS3** following design-system principles, integrated **Generative AI** capabilities to summarize insights and recommend actions, and shipped features on quarterly release cadence achieving **98% user adoption** within first 90 days.
- Implemented comprehensive **test-driven development** and **Software Engineering processes** including automated testing suites, peer code reviews, and **CI/CD pipelines with Jenkins**; improved defect detection by **55%** pre-production while maintaining **99.7% application availability**.
- Integrated platform with existing enterprise business systems via **REST APIs**, enabling seamless data flow across **relational databases** and reducing manual data reconciliation overhead by **70%**; documented architecture and workflows enabling rapid onboarding of 8 new development teams.

SWIFT Payments Operations Dashboard & ISO 20022 Migration Portal (HCL Technologies)

- Engineered production-grade payments operations dashboard using **Power BI** and **Tableau** for banking domain, implementing **DAX measures** and **Power Query** ETL workflows to deliver real-time transaction visibility across SWIFT messaging infrastructure; supported **99.99% SLA** for payment processing with sub-second alerting on P1/P2 incidents.
- Designed **RESTful API layer in Java Spring Boot** and **Python** integrating payment systems with **relational databases (PostgreSQL, Oracle)**, translating **ISO 20022 message flows** and **SEPA/cross-border payment** workflows into actionable operational metrics; reduced transaction reconciliation time by **65%**.
- Partnered with banking domain stakeholders, business analysts, and regulatory teams to understand compliance requirements (GDPR, PSD2, Open Banking), championing platform enhancements that ensured audit-trail completeness and regulatory reporting accuracy across interconnected enterprise systems.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)